



A+ LAB SERIES V5

Lab 1B: Examining PC Hardware

Document Version: **2026-01-02**

Material in this Lab Aligns to the Following	
CompTIA A+ (220-1201) Exam Objectives	3.5: Given a scenario, install and configure motherboards, central processing units (CPUs), and add-on cards. 5.2: Given a scenario, troubleshoot motherboards, RAM, CPUs, and power. 5.3: Given a scenario, troubleshoot video, projector, and display issues.
CompTIA A+ (220-1202) Exam Objectives	1.3: Compare and contrast basic features of Microsoft Windows editions. 1.4: Given a scenario, use Microsoft Windows operating system features and tools. 1.5: Given a scenario, use the appropriate Microsoft command-line tools. 4.1: Given a scenario, implement best practices associated with documentation and support systems information management.

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Introduction

Understanding the intricacies of computer hardware components and their configurations is fundamental not only for computer technicians but also for everyday users. A solid grasp of this knowledge empowers individuals to manage and maintain their computers more effectively, ensuring they perform optimally and can be easily upgraded or repaired when necessary.

This lab will provide a comprehensive, hands-on approach to exploring the various hardware components that constitute a computer system. Through the use of Windows-based tools, such as the Control Panel and Command Prompt, along with specialized utilities like MSinfo32.exe and DxDiag, you will delve into the detailed specifications and status of critical components including the processor (CPU), memory (RAM), storage devices (hard drives, SSDs), network adapters, and more.

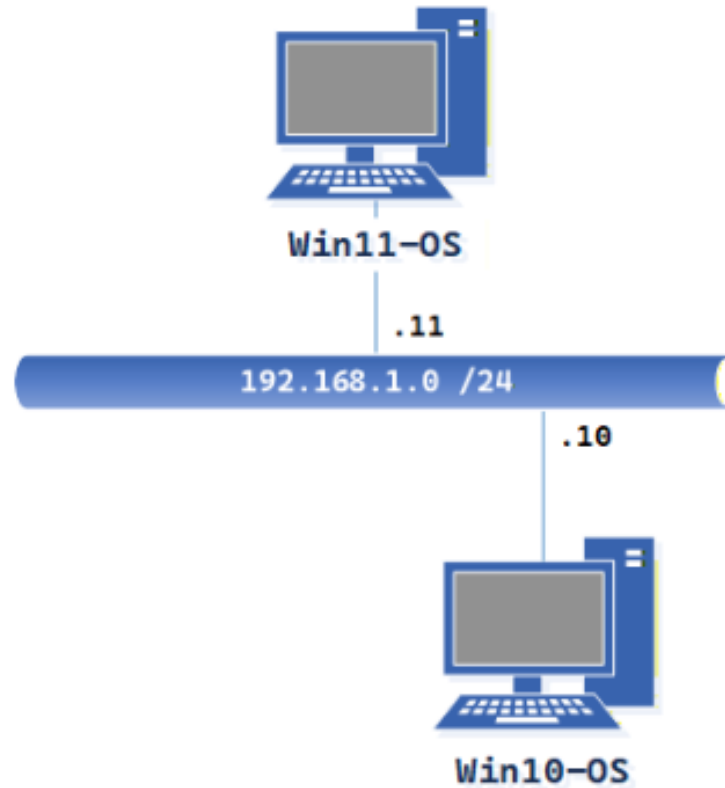
Each tool and utility offers a unique perspective on the system's hardware, from general overviews to in-depth technical details. By familiarizing yourself with these methods, you will not only enhance your ability to troubleshoot and resolve hardware-related issues but also optimize your system's performance. Furthermore, this knowledge will enable you to make well-informed decisions when considering hardware upgrades or replacements, ensuring you can adapt your computer system to meet evolving demands and technologies.

In a world where technology is constantly advancing, the ability to understand and interact with your computer's hardware is an invaluable skill. Whether you're a technician ensuring systems run smoothly or a user aiming to get the most out of your device, mastering these tools and techniques will place you in control of your computing environment.

Objective

In this lab, you will examine computer system hardware using tools that are inherent in *Windows*-based computers. You will also explore a third-party tool. These tools and utilities help identify computer hardware components and their configurations. Additionally, you will have the opportunity to review key concepts related to computer hardware and system utilities.

Lab Topology



Lab Settings

The information in the table below will be needed to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Win10-OS	192.168.1.10 / 24	sysadmin	NDGLabpass123!
Win11-OS	192.168.1.11 / 24	sysadmin	NDGLabpass123!

1 Accessing System Information

The **Control Panel** in Windows 10 and 11 is a centralized location for accessing and managing system settings. It provides a range of options for configuring hardware, software, network settings, and other aspects of the operating system.

Please
Note

There are some differences between the Control Panel in Windows 10 and Windows 11. While both operating systems offer the Control Panel for managing system settings, Windows 11 has gradually shifted towards the Settings app as the primary interface for configuration.

Here are the key differences:

- **Accessibility:** In Windows 11, the Control Panel is less prominently displayed and can be found by searching for it in the Start menu. Windows 10 places the Control Panel more prominently on the Start menu.
- **Functionality:** While the Control Panel in both Windows 10 and Windows 11 offers similar functionalities, Windows 11 has gradually incorporated more settings into the Settings app, making it a more comprehensive tool for managing system configurations.
- **Design:** The Control Panel in Windows 11 maintains a more traditional look and feel compared to the redesigned Settings app in Windows 11.

In summary, while the Control Panel remains a valuable tool in Windows 11 for accessing certain system settings, especially for legacy applications or advanced users, Windows 11 has increasingly focused on the Settings app as the primary interface for managing system configurations.

1.1 System Control Panel Applet in Windows 10

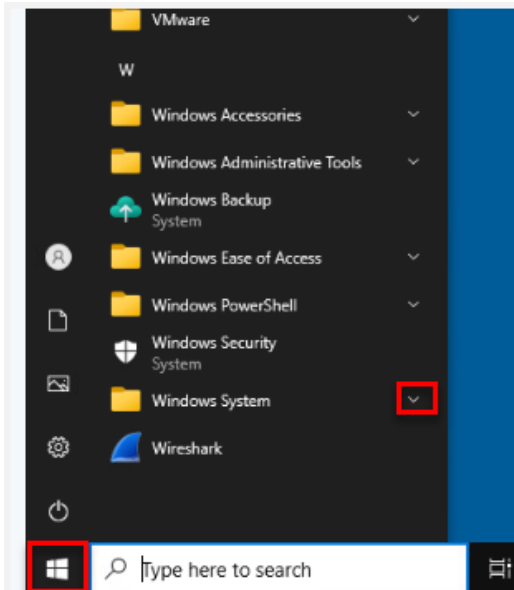
1. Open a console connection to the **Win10-OS** system.



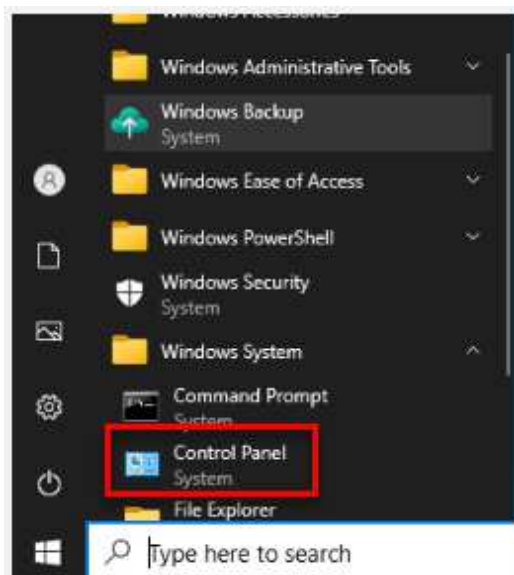


To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

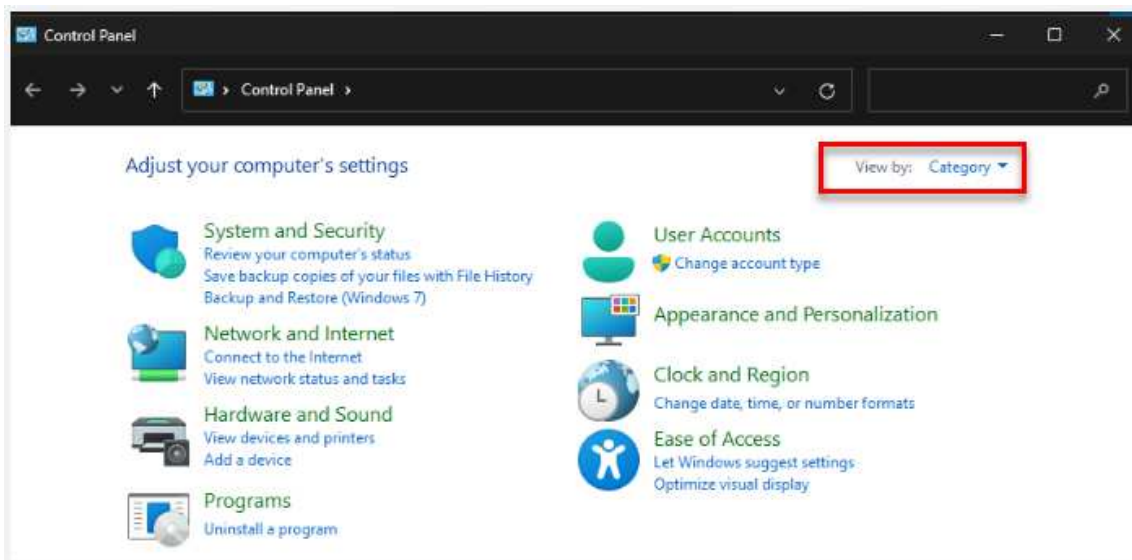
- Click on the **Start** button and then scroll through the apps and select the drop-down arrow of the **Windows System** folder.



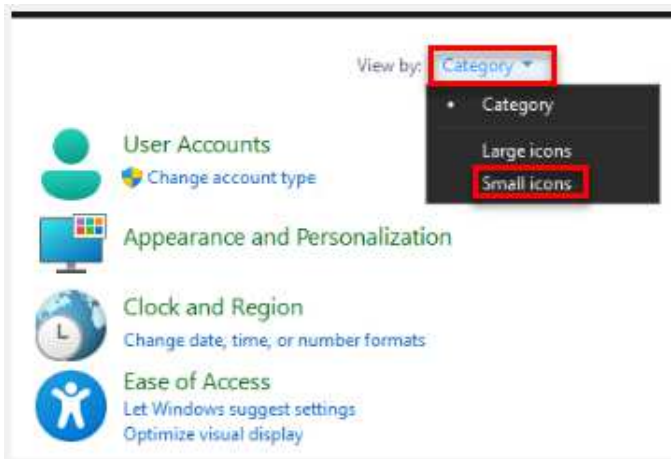
- Select **Control Panel (System)**.



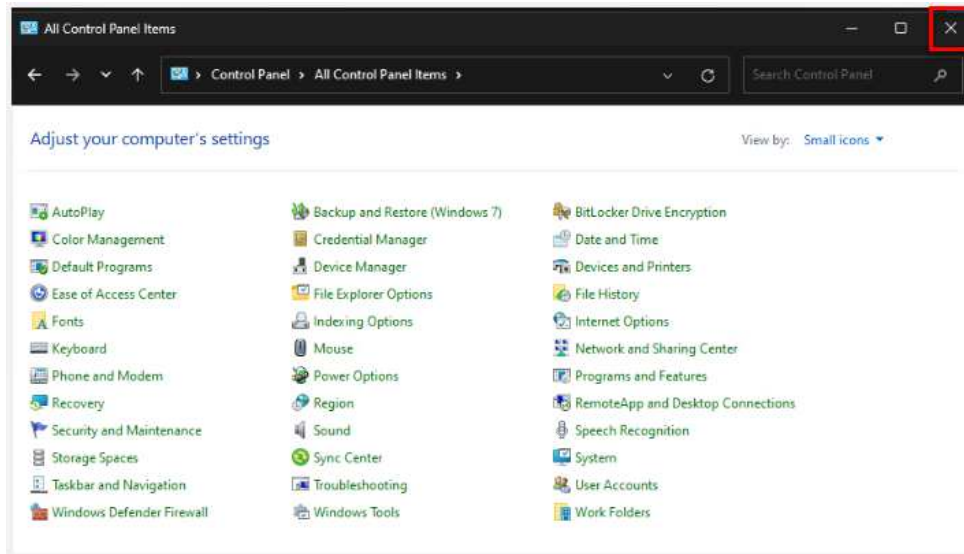
4. Review the Computer Settings, which are Adjustable under the View by: **Category** Section.



5. Select the drop-down arrow next to **Category** and select **Small Icons**.



- Review the Computer Settings, which are now available under the **Small icons** of the *Control Panel*, and options to adjust your computer's settings.



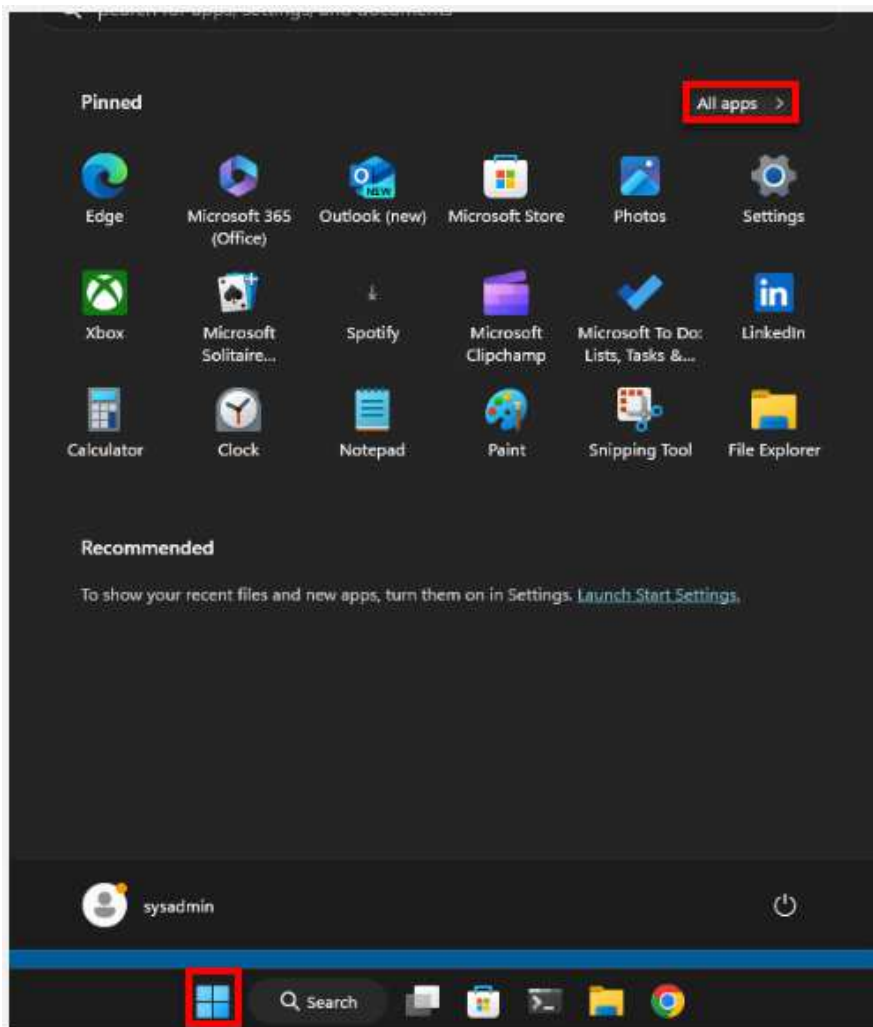
- Close each remaining window by clicking the **X** in the top-right corner.

1.2 System Control Panel Applet in Windows 11

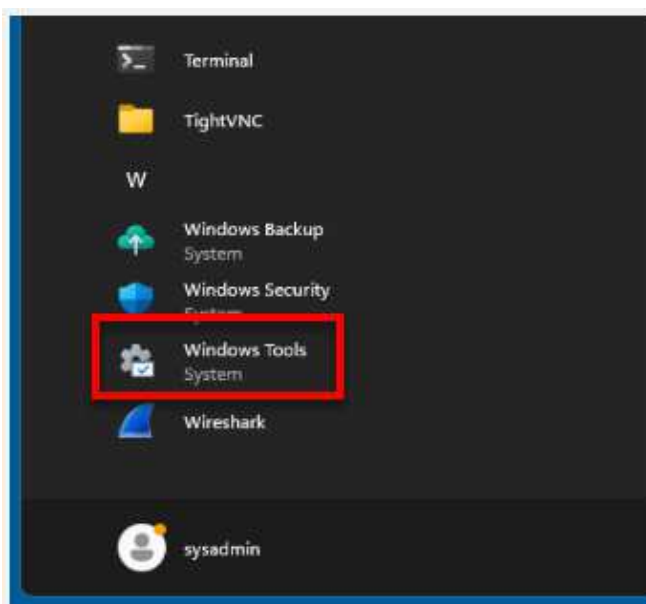
- Open a console connection to the **Win11-OS** system.



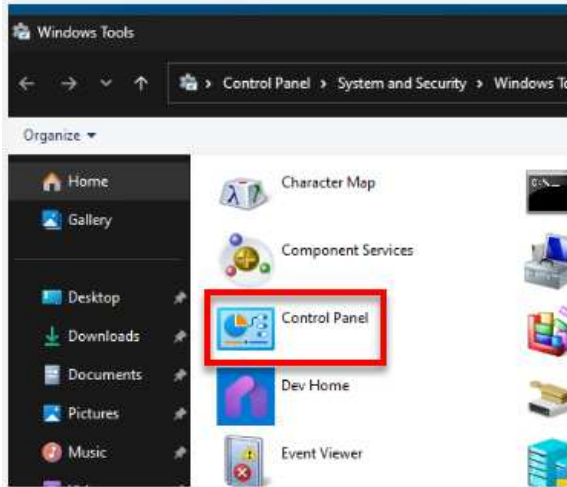
- Click on the **Start** button and then select **All apps**.



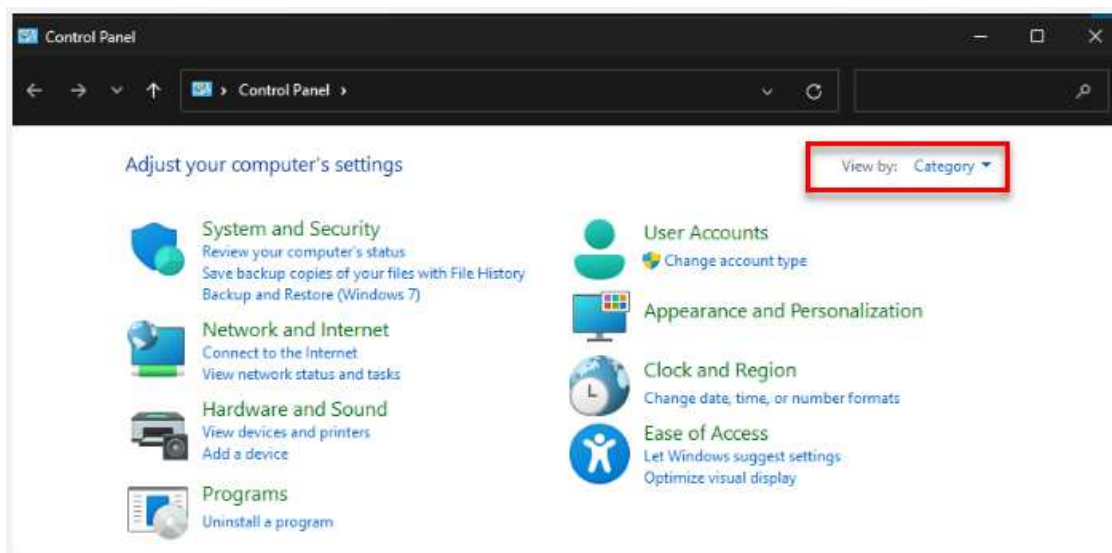
- Scroll through the *All Apps* options and select **Windows Tools (System)**.



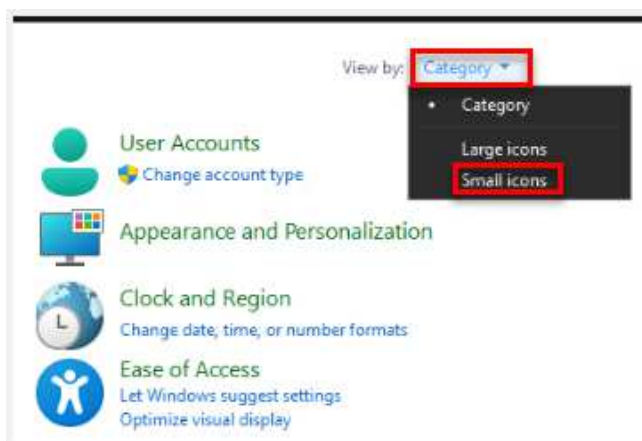
4. Double-click the **Control Panel** icon.



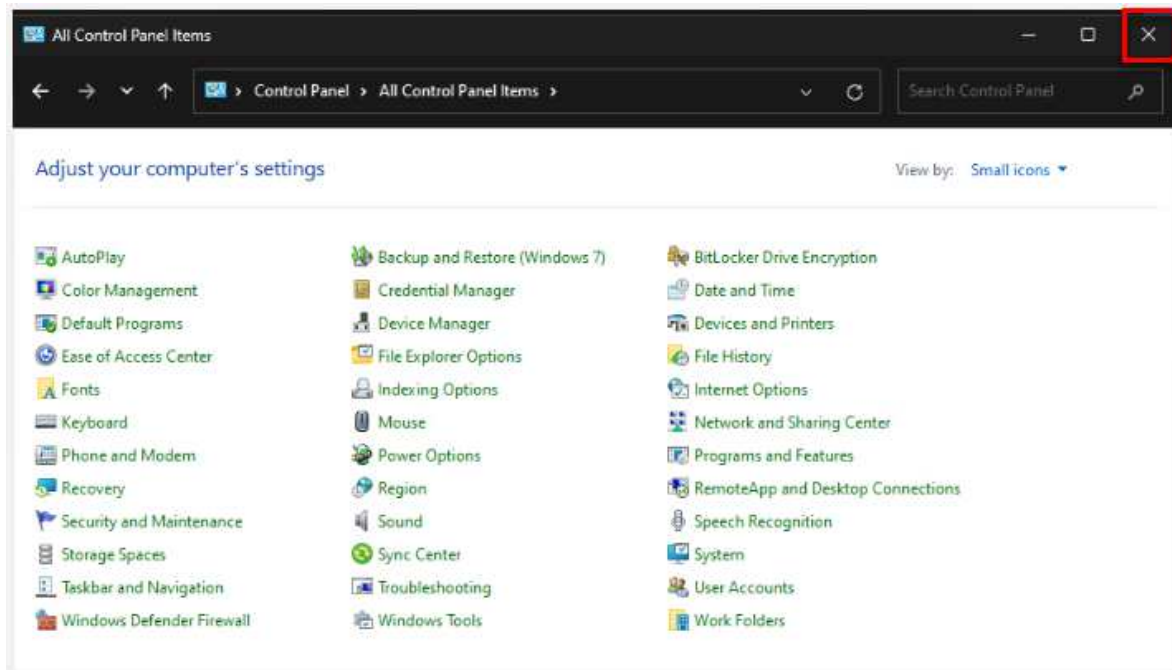
5. Review the Computer Settings, which are Adjustable under the View by **Category** Section.



6. Select the drop-down arrow next to Category and select **Small Icons**.



- Review the Computer Settings, which are now available under the **Small icons** of the Control Panel, and options to adjust your computer's settings.



- Close each remaining window by clicking the **X** in the top-right corner.

2 Accessing Systeminfo.exe

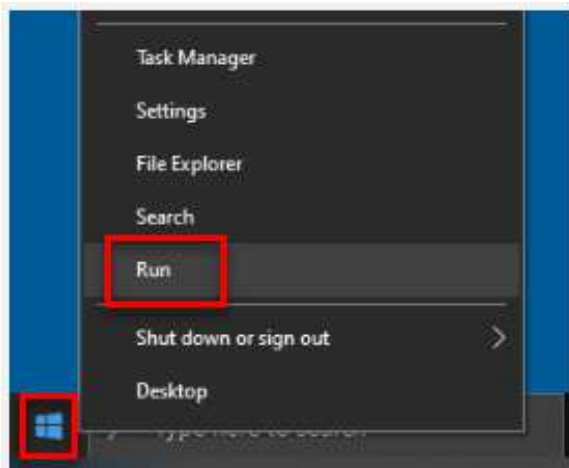
Systeminfo.exe is a valuable tool for gathering detailed system information in Windows. It can be used to troubleshoot issues, understand system performance, and gather data for technical support. By using systeminfo.exe effectively, you can gain valuable insights into your computer's hardware and software, helping you troubleshoot problems, optimize performance, and make informed decisions about system maintenance.

2.1 Accessing Systeminfo.exe in Windows 10

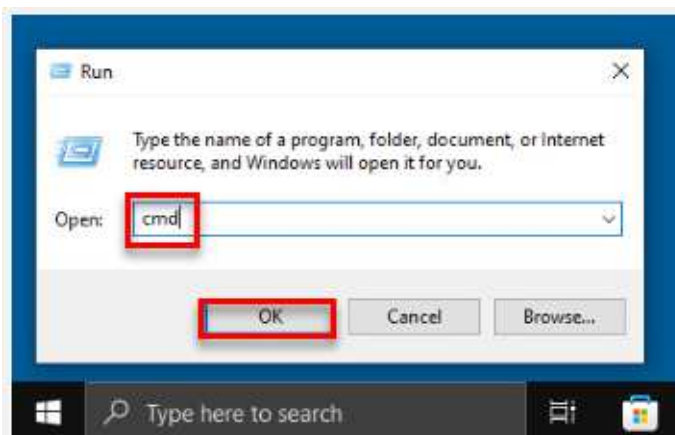
1. Switch to the **Win10-OS** system.



2. In the lower-left corner of the screen, right-click on the **Start** button. The *Power User Menu* displays, click on **Run**.

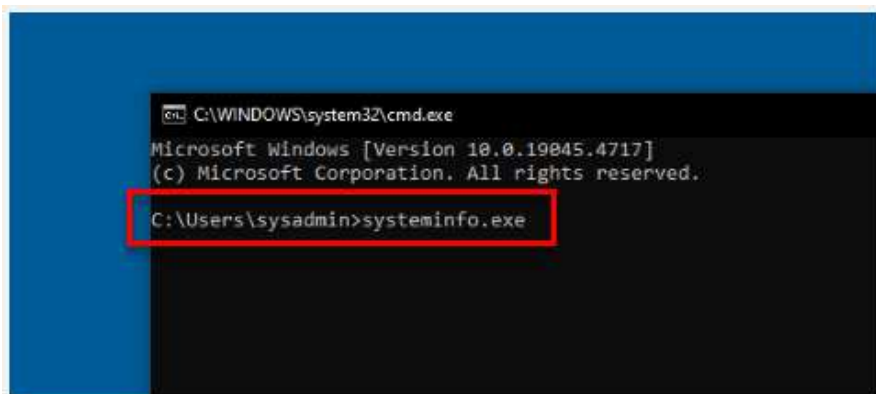


3. Type **cmd** in the text field and press the **Enter** key or click on **OK**.



4. In the *Command Prompt* window, type `systeminfo.exe` at the prompt and press **Enter** to run the utility.

```
systeminfo.exe
```



5. In the *Command Prompt* window, scroll through and review the information provided by the utility.

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.4717]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sysadmin>systeminfo.exe

Host Name:                WIN10-OS
OS Name:                  Microsoft Windows 10 Pro
OS Version:               10.0.19045 N/A Build 19045
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:         sysadmin
Registered Organization:   N/A
Product ID:               00331-20473-07490-AA287
Original Install Date:     3/23/2023, 5:04:11 AM
System Boot Time:          8/31/2024, 2:54:06 PM
System Manufacturer:       VMware, Inc.
System Model:              VMware Virtual Platform
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 85 Stepping 4 GenuineIntel ~2295 Mhz
BIOS Version:              Phoenix Technologies LTD 6.00, 11/12/2020
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:              en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC-05:00) Eastern Time (US & Canada)
Total Physical Memory:      4,095 MB
Available Physical Memory:  2,166 MB
Virtual Memory: Max Size:  4,799 MB
Virtual Memory: Available: 3,062 MB
Virtual Memory: In Use:     1,737 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:               \\WIN10-OS
```

6. Close the *command prompt* by typing **exit** followed by pressing the **Enter** key.

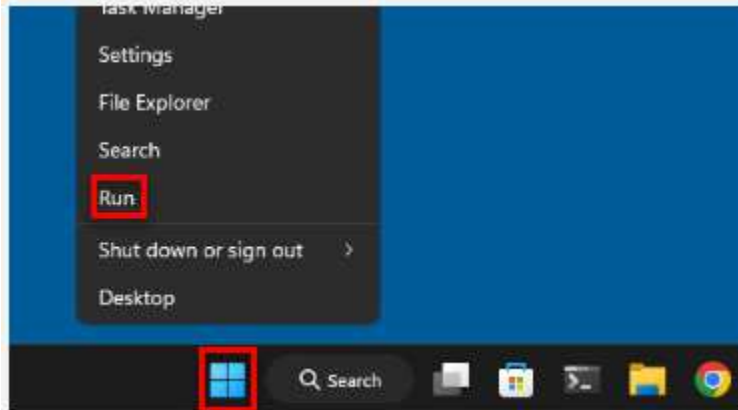
```
C:\Users\sysadmin>exit
```

2.2 Accessing systeminfo.exe in Windows 11

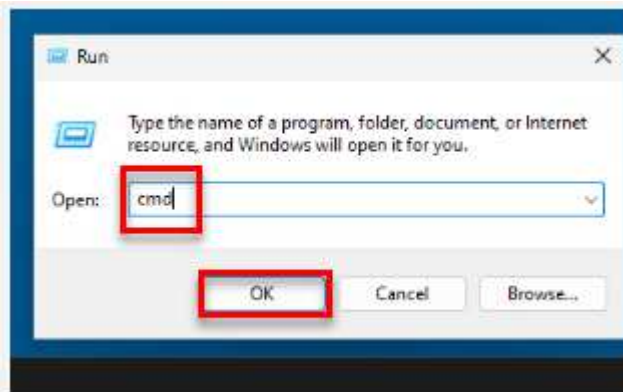
1. Switch to the **Win11-OS** system.



- In the taskbar at the bottom of the screen, right-click on the **Start** button. The *Power User Menu* displays, click on **Run**.

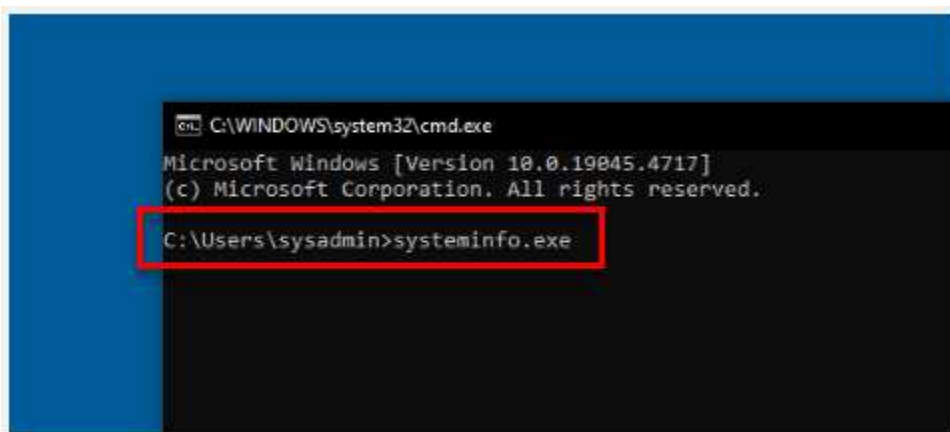


- Type `cmd` in the text field and press the **Enter** key or click on **OK**.

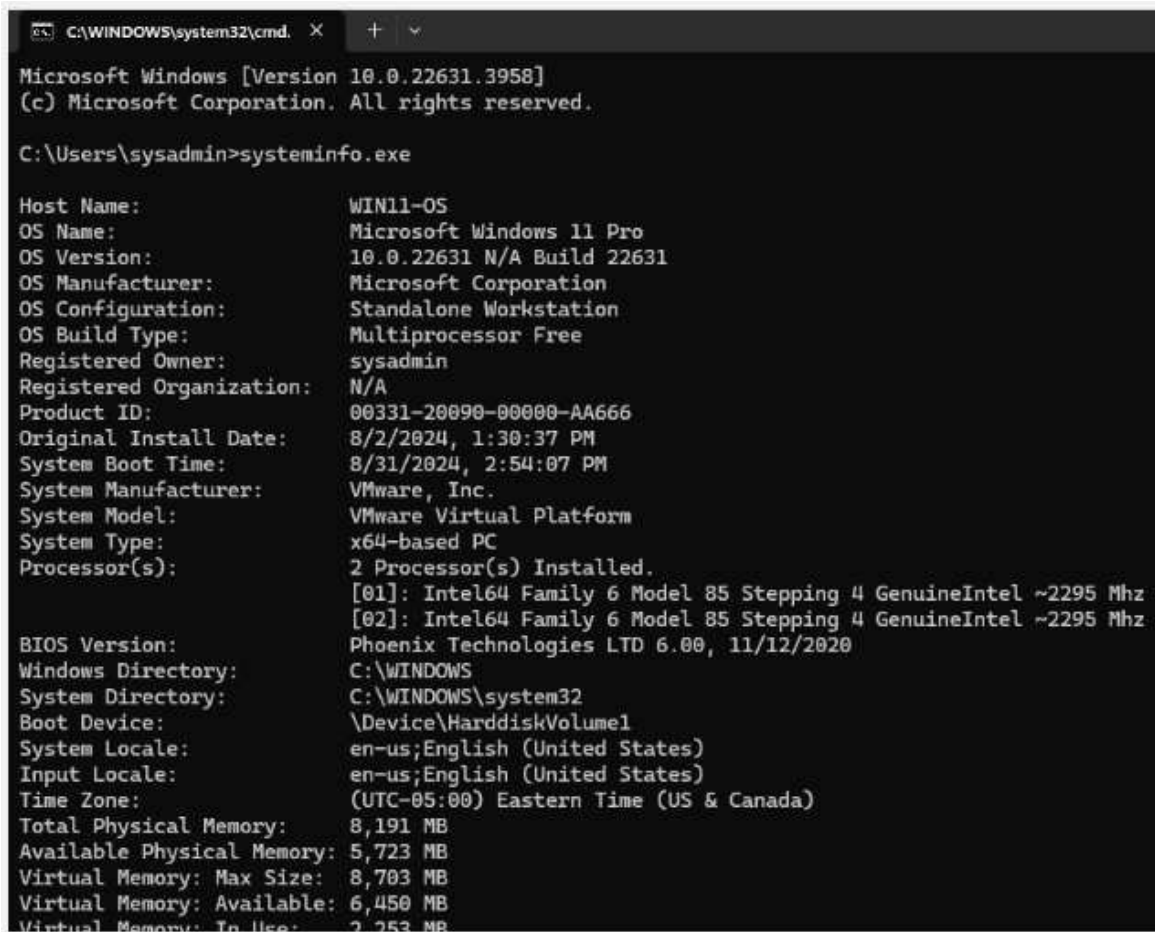


- In the *Command Prompt* window, type `systeminfo.exe` at the prompt and press **Enter** to run the utility.

```
systeminfo.exe
```



5. In the *Command Prompt* window, scroll through and review the information provided by the utility.

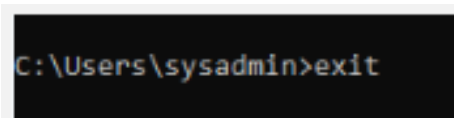


```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.22631.3958]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sysadmin>systeminfo.exe

Host Name:                               WIN11-OS
OS Name:                                 Microsoft Windows 11 Pro
OS Version:                             10.0.22631 N/A Build 22631
OS Manufacturer:                       Microsoft Corporation
OS Configuration:                      Standalone Workstation
OS Build Type:                           Multiprocessor Free
Registered Owner:                       sysadmin
Registered Organization:                N/A
Product ID:                             00331-20090-00000-AA666
Original Install Date:                  8/2/2024, 1:30:37 PM
System Boot Time:                      8/31/2024, 2:54:07 PM
System Manufacturer:                   VMware, Inc.
System Model:                           VMware Virtual Platform
System Type:                            x64-based PC
Processor(s):                           2 Processor(s) Installed.
[01]: Intel64 Family 6 Model 85 Stepping 4 GenuineIntel ~2295 Mhz
[02]: Intel64 Family 6 Model 85 Stepping 4 GenuineIntel ~2295 Mhz
BIOS Version:                           Phoenix Technologies LTD 6.00, 11/12/2020
Windows Directory:                     C:\WINDOWS
System Directory:                       C:\WINDOWS\system32
Boot Device:                            \Device\HarddiskVolume1
System Locale:                           en-us;English (United States)
Input Locale:                           en-us;English (United States)
Time Zone:                              (UTC-05:00) Eastern Time (US & Canada)
Total Physical Memory:                  8,191 MB
Available Physical Memory:              5,723 MB
Virtual Memory: Max Size:               8,703 MB
Virtual Memory: Available:              6,450 MB
Virtual Memory: In Use:                  2,253 MB
```

- 3 Close the *command prompt* by typing `exit`, followed by pressing the **Enter** key.



```
C:\Users\sysadmin>exit
```

- 4 Leave the **Win11-OS** machine open to continue with the next task.

3 Accessing and Working with the MSInfo32 Utility

The *MSinfo32 Utility* (msinfo32.exe) is a powerful diagnostic tool built into Windows that provides a detailed overview of a computer's hardware configuration, components, and software environment, including drivers. This utility is an essential resource for both IT professionals and advanced users, offering critical insights into the system's inner workings. Whether you're troubleshooting issues, planning hardware upgrades, or simply gaining a deeper understanding of your system's configuration, MSInfo32 is invaluable.

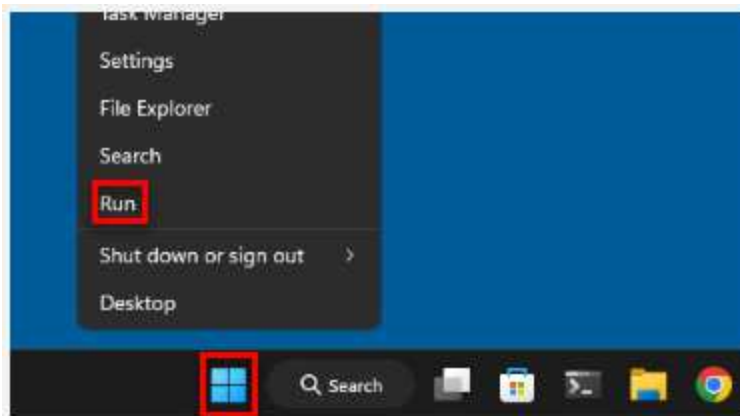
System Information lists categories in the left pane and details about each category in the right pane. The categories include:

1. *System Summary* - Displays general information about your computer and the operating system, such as the computer name and manufacturer, the type of basic input/output system (BIOS) your computer uses, and the amount of memory that's installed.
2. *Hardware Resources* - Displays advanced details about your computer's hardware and is intended for IT professionals.
3. *Components* - Displays information about disk drives, sound devices, modems, and other components installed on your computer.
4. *Software Environment* - Displays information about drivers, network connections, and other program-related details.

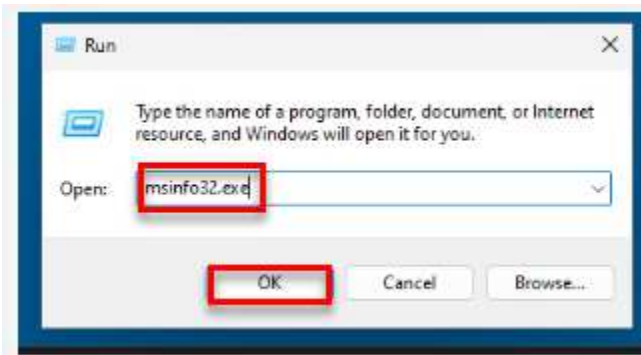
In this exercise, you will explore using the *System Information* or *MSInfo32 Utility*. Choose any Windows host and launch it if you are not already running one.

3.1 Accessing the MSInfo32 Utility

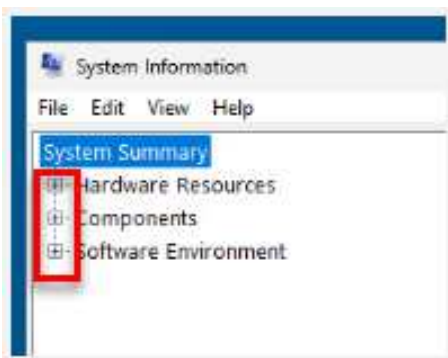
1. While on the *Win11-OS* system, in the taskbar at the bottom of the screen, right-click on the **Start** button. The *Power User Menu* displays, click on **Run**.



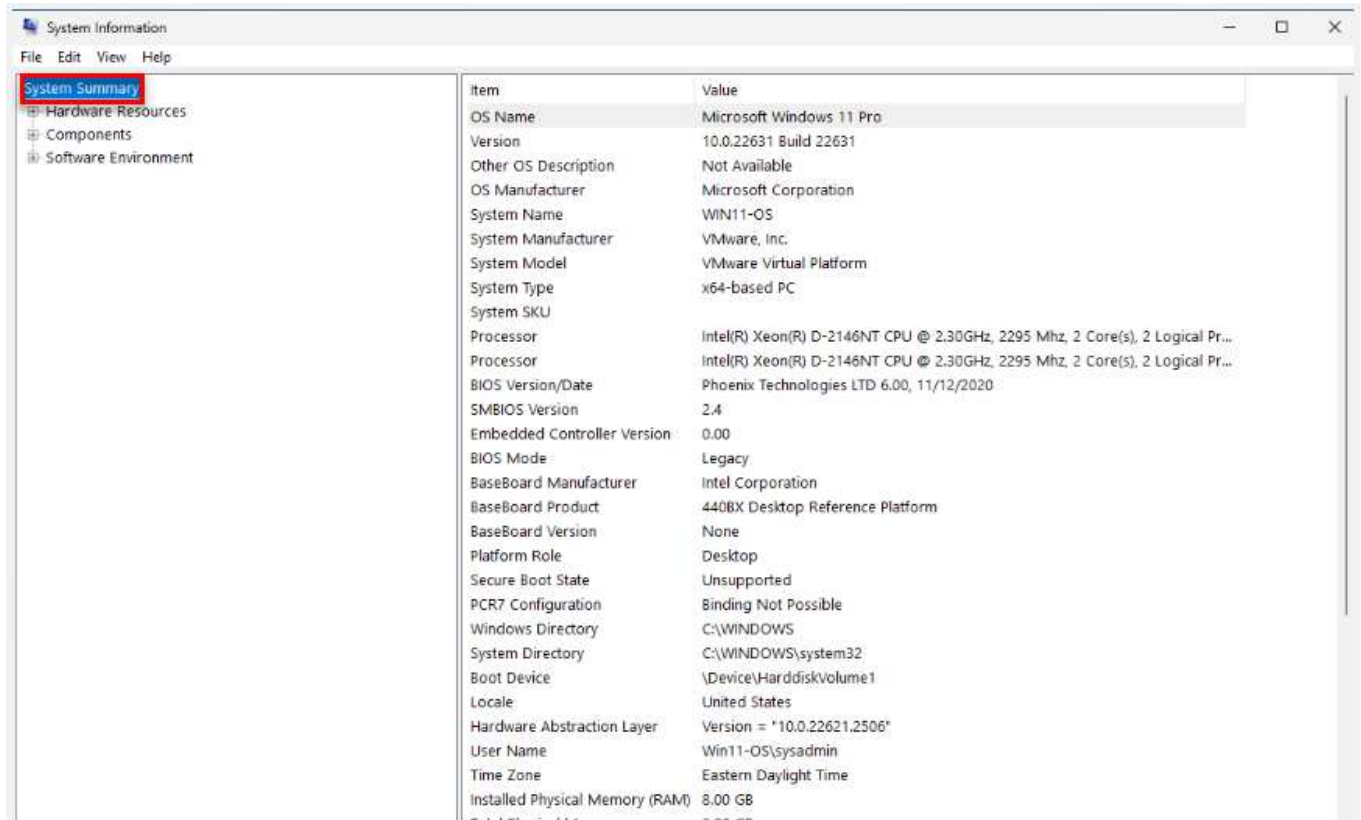
2. In the *Command Prompt* window, type `msinfo32.exe` at the prompt and press **Enter** or select **OK** to run the utility.



3. The *System Information* window appears. In the left panel, click the + sign next to each category to see what information is available in this utility.
- 4.

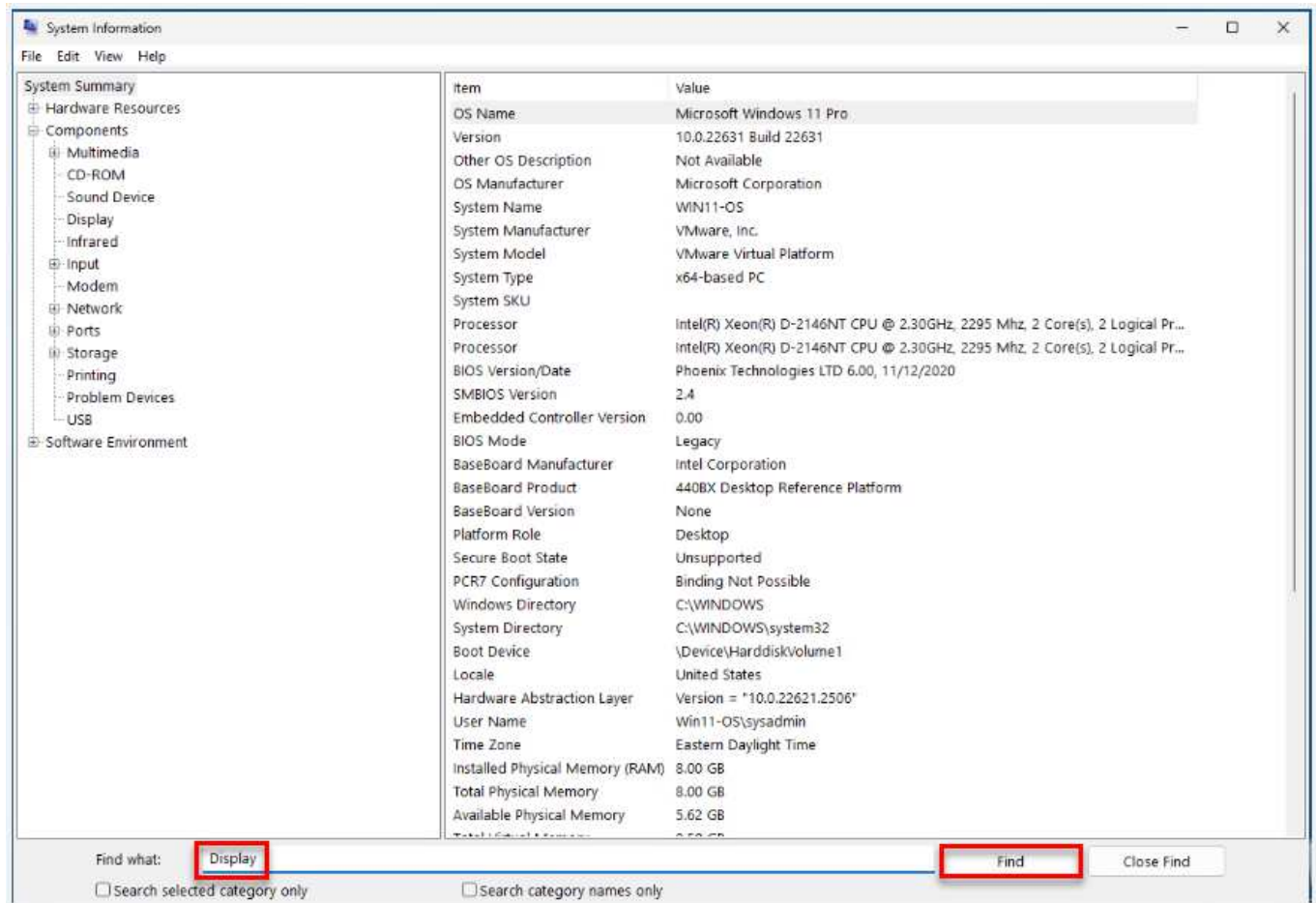


- Go back to the top left panel, click, and highlight **System Summary**. Inspect the information in the right pane. This includes information on the system model, system type, processor, memory, and BIOS information.

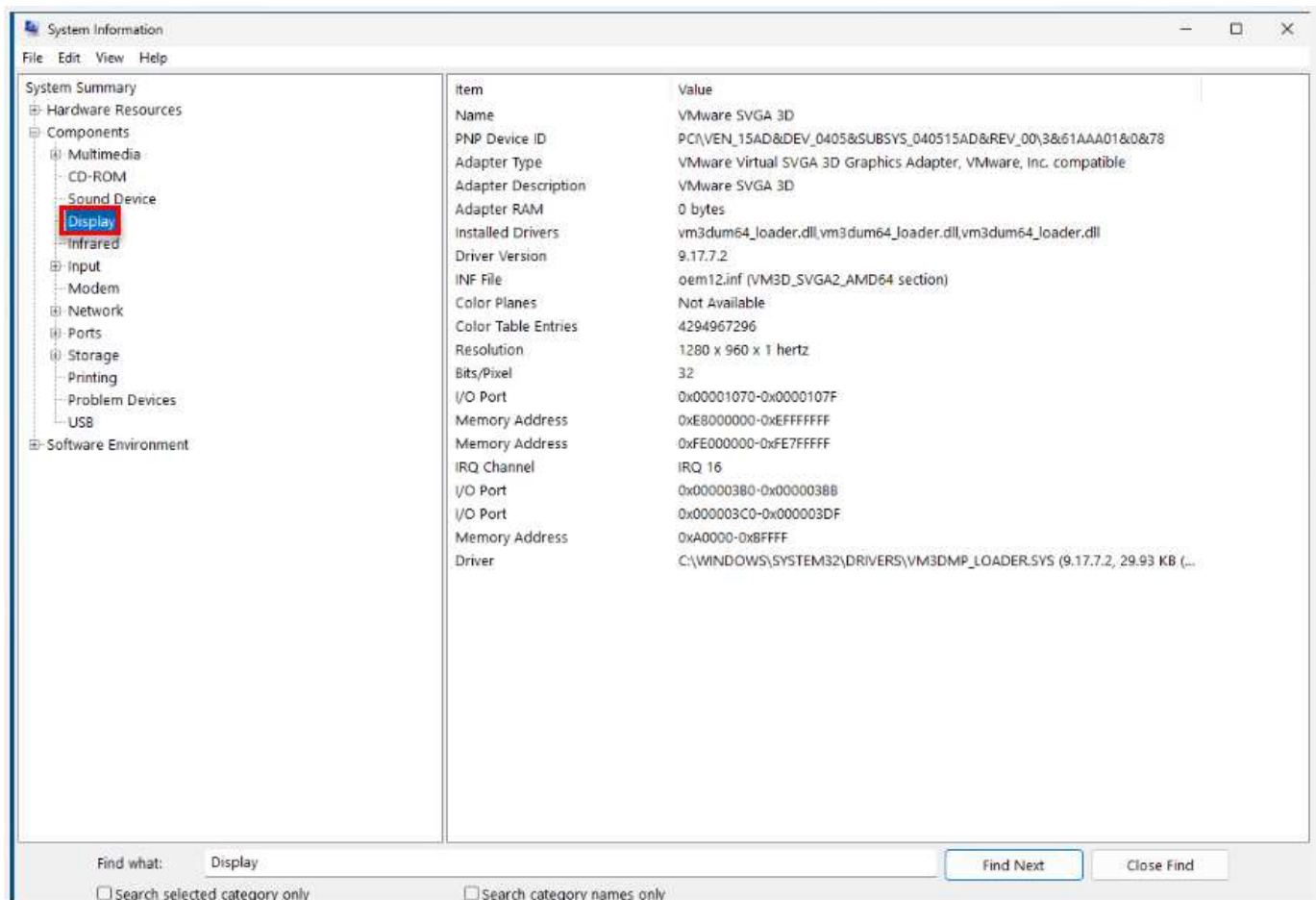


Not all hardware information is located on the *System Summary*. To locate information about specific components, click on the + sign next to the *Components* category, or use the find feature. It is necessary to be specific in providing information, as the search is not intuitive.

6. Type **Display** into the *Find what:* text field found at the bottom of the window. Click the **Find** button. (It may be required to click **Find Next** until you get to the specific information you wish to obtain.)



7. Notice the search feature found the *Display* subcategory underneath the *Components* category and helped identify information about the display adapter.
- 8.



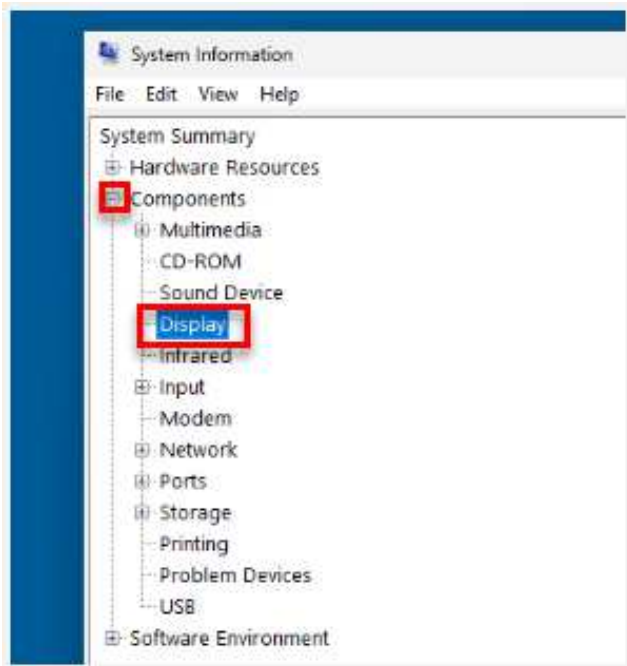
9. Leave the **Win11-OS** machine open to continue with the next task.

3.2 Generating a System Information Report

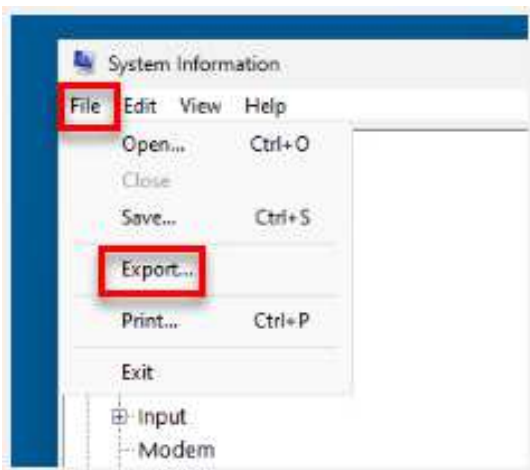
Creating and saving reports about your system specifications is a handy feature. Create a baseline of the system when new, then again at each change so that if troubleshooting is necessary, your reports are ready for reference, especially when a system will not boot. This exercise will have you generate a report.

1. While on the **Win11-OS** machine, if not already opened, open **System Information** (*MSinfo32 Utility*).
2. Select the information for which you wish to create a report. In this example, you will use *Display*. This would be useful to do before you update the display adapter drivers.

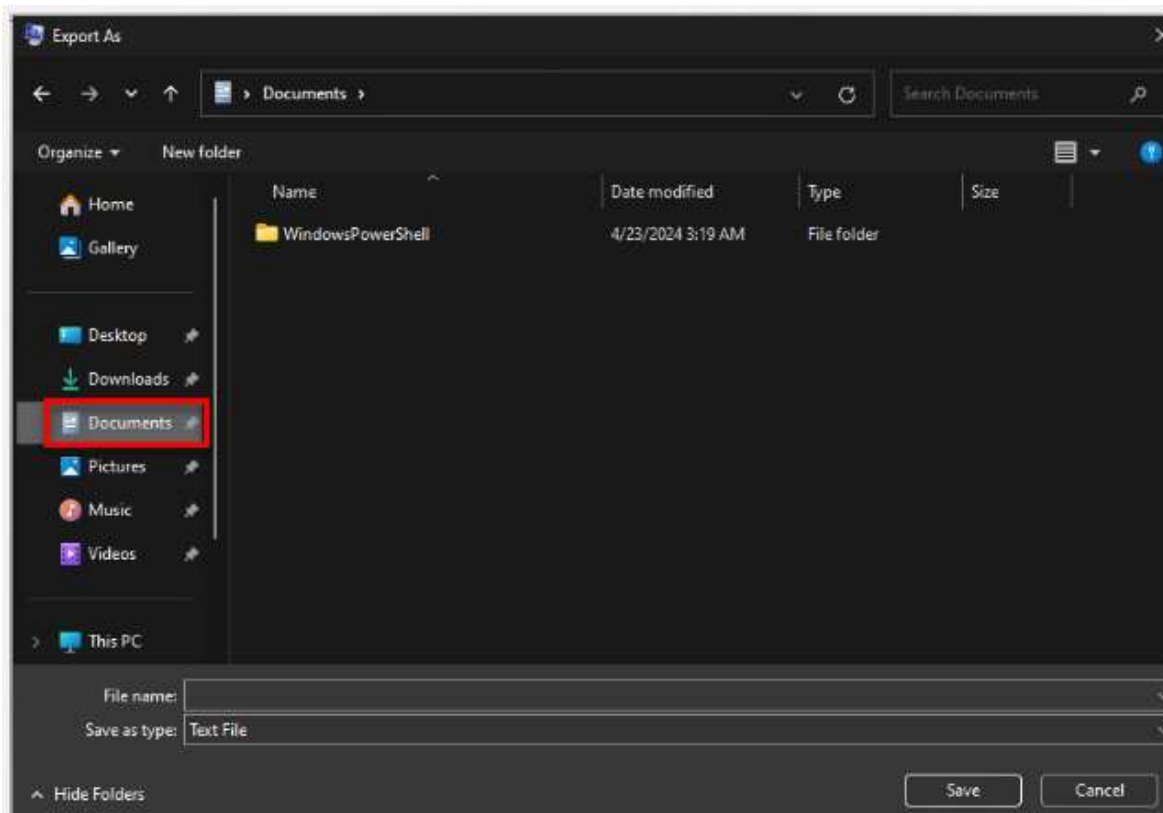
3. If *Display* is not selected, click on the + sign next to the **Components** category and select **Display**. This will put the *Display* adapter information in the right pane.



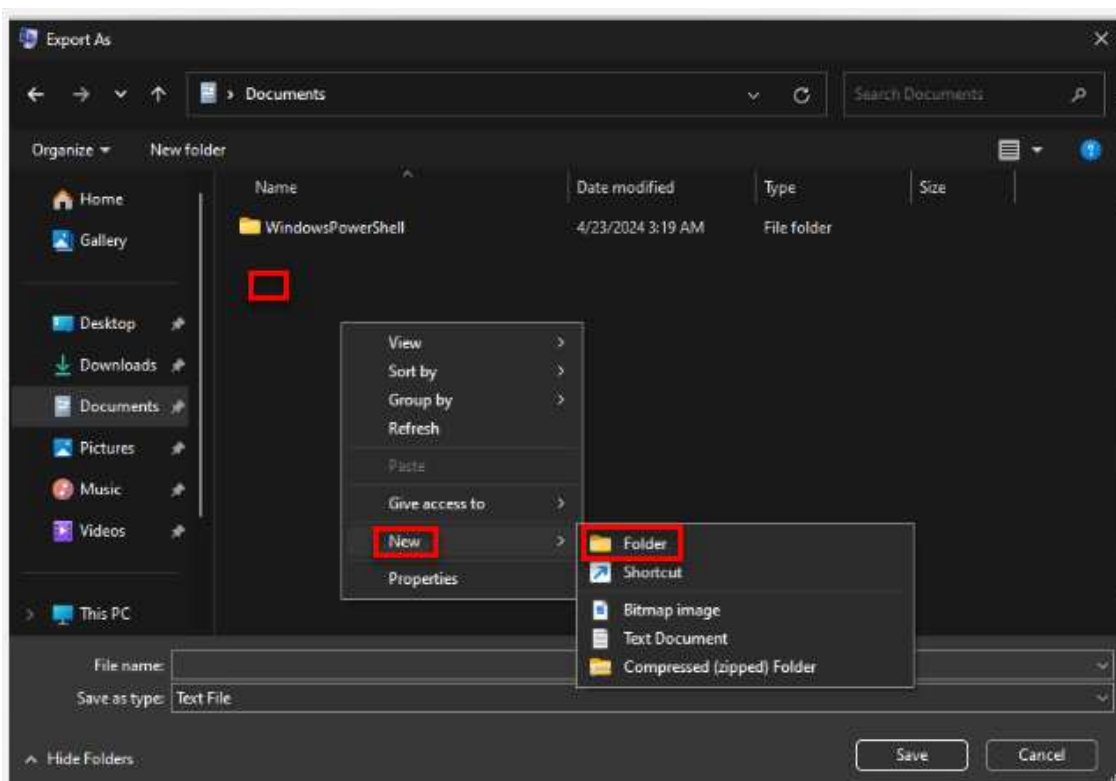
4. Click on **File > Export** in the menu bar.



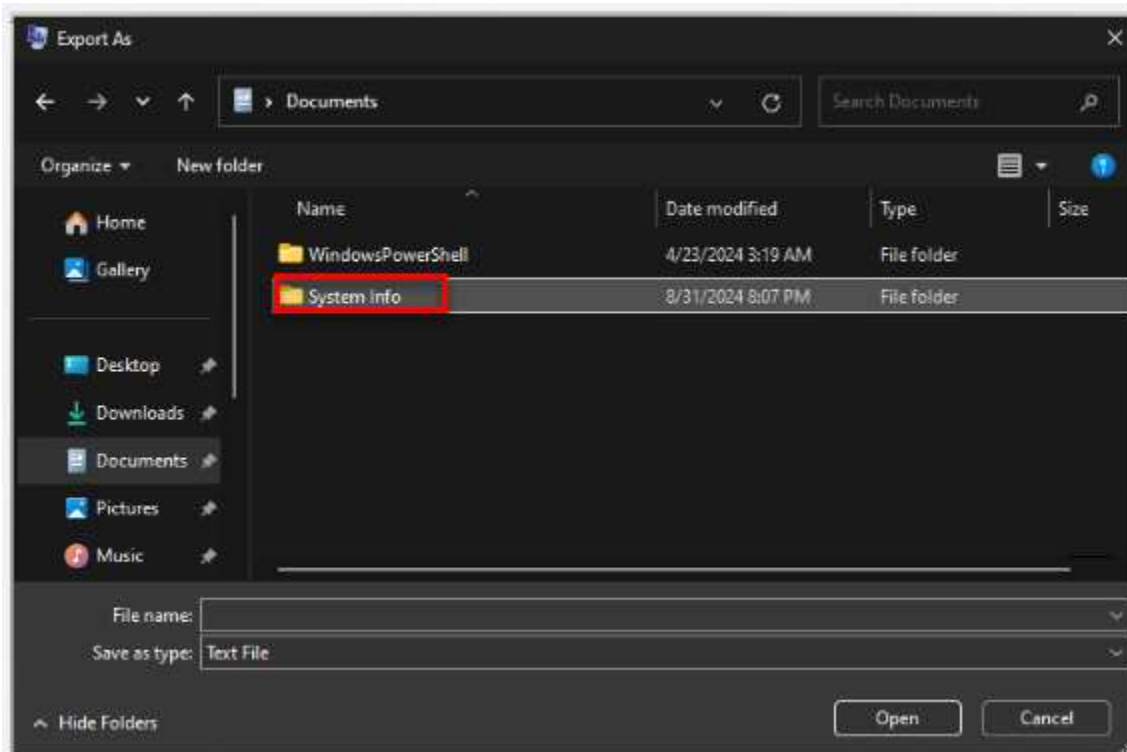
5. The *Export As* dialog box appears. Navigate to the **Documents** folder from the *Quick access* pane.



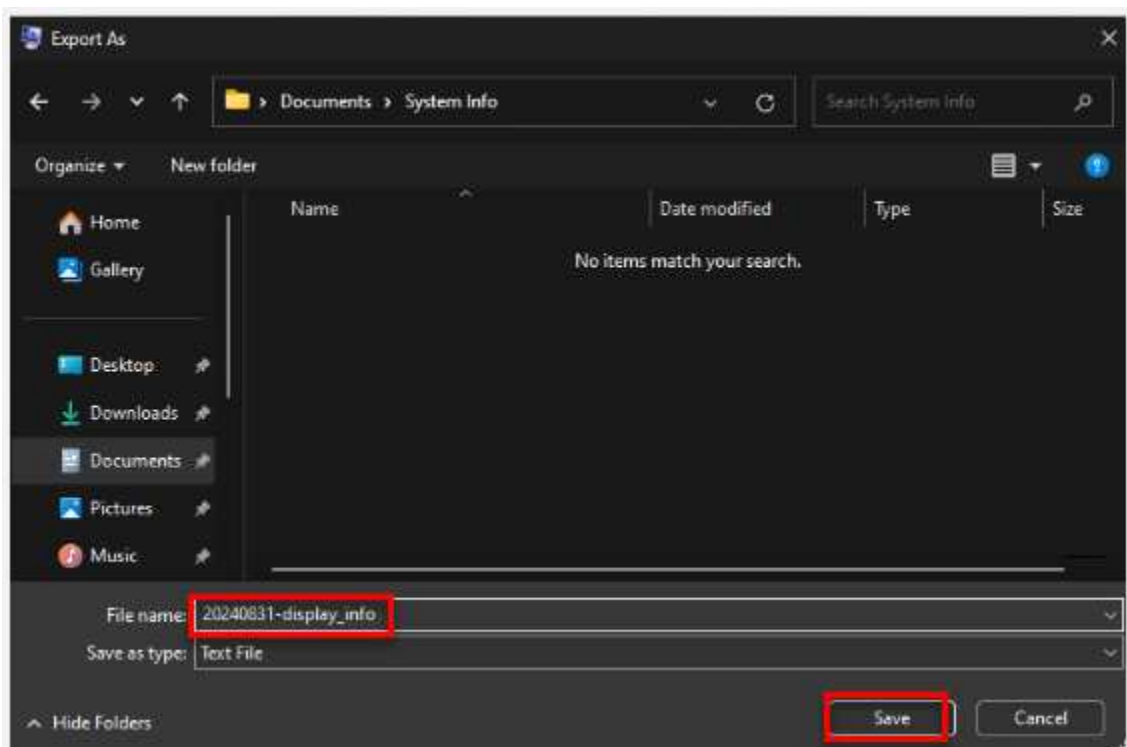
6. In the *Export As* window, right-click in the blank space under *WindowsPowerShell*, and select **New > Folder**.



7. When naming the new folder, type **System Info**, and press **Enter**. Once created, double-click on the **System Info** folder.



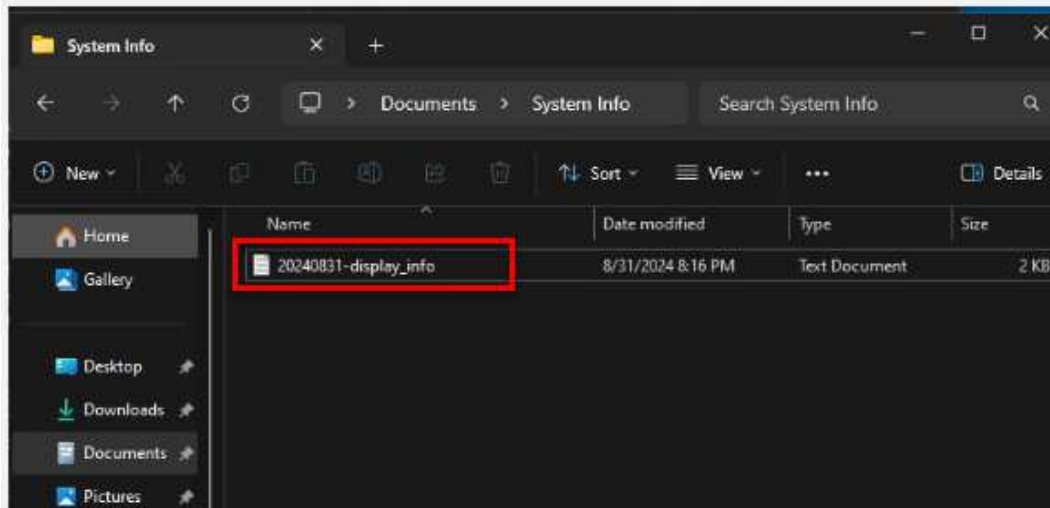
8. Type `<date>-display_info` into the *File name* text field and click **Save**. (For this example, "20240831-display_info", was used as the filename.)



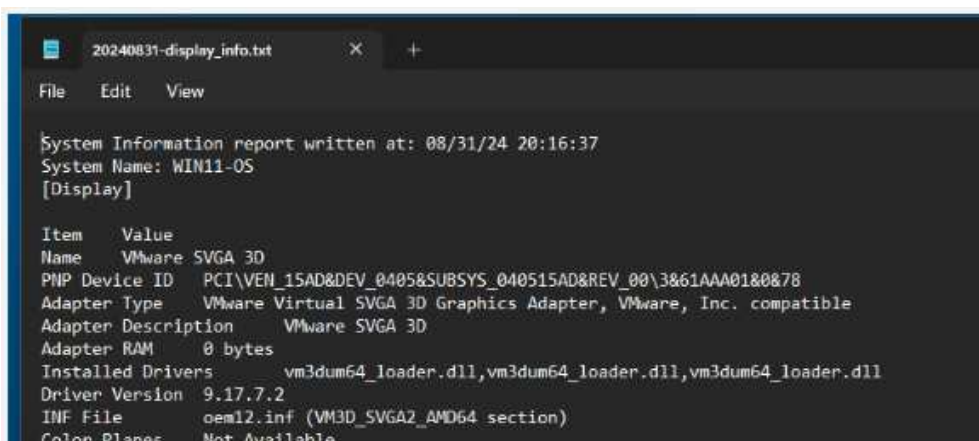
9. Close the *System Information* window by clicking the **X** in the upper-right corner.
10. Open a **File Explorer** window by clicking on the icon located in the taskbar.



11. In the *File Explorer* window, navigate to the **Documents > System Info**.
12. Double-click the file you just created from the export process.



13. Notice the report displays the text file with the display adapter information.



14. Close each remaining window by clicking the **X** in the top-right corner.
15. Leave the *Win11* machine open to continue with the next task.



A best practice would be to save this file externally in case the system cannot boot. If needed, you would then be able to retrieve the information.

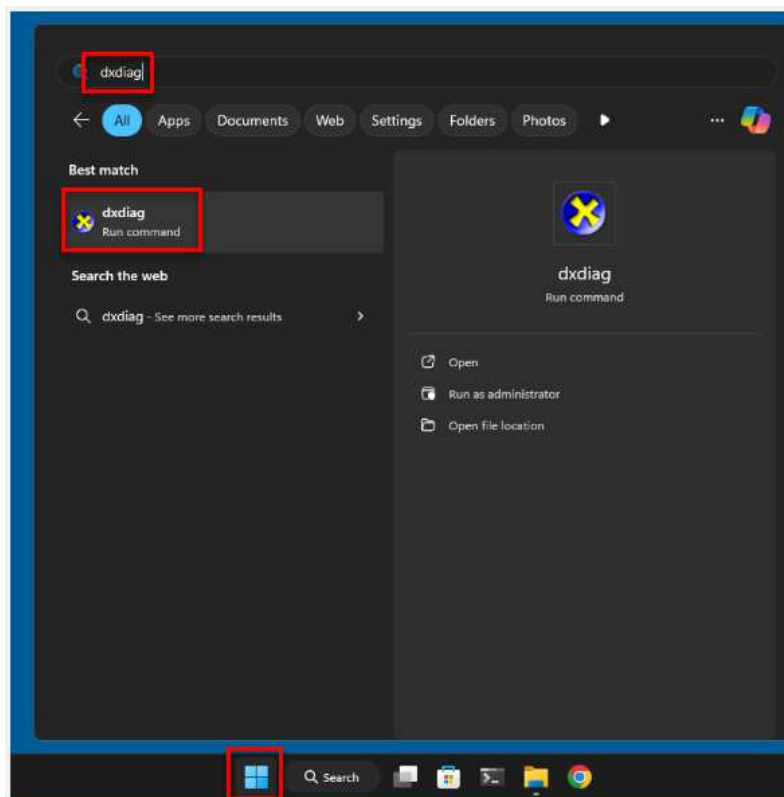
4 DirectX Diagnostic Tool

The DirectX Diagnostic Tool (dxdiag.exe) is a crucial utility built into Windows, designed to help users diagnose and troubleshoot DirectX-related issues, which is essential for multimedia tasks such as gaming and video playback. This tool provides detailed information about your system's graphics adapter, sound card, and input devices, making it a valuable resource for ensuring the smooth performance of multimedia applications. DirectX functions as an interface between your PC's hardware and the Windows operating system, enabling efficient operation of games and other multimedia applications. As these applications have become increasingly complex and demanding, the ability to identify and resolve DirectX-related problems has become more important than ever.

The DirectX Diagnostic Tool allows users and technicians to diagnose hardware issues, verify driver compatibility, test DirectX features such as DirectDraw and Direct3D, and gather detailed system information for troubleshooting or technical support. Notably, the functionality of dxdiag.exe remains consistent between Windows 10 and Windows 11. Both versions of the tool offer the same essential features, enabling users to check device drivers, test DirectX capabilities, and similarly gather system information across both operating systems. While there may be minor visual differences or interface tweaks between the two versions, these do not impact the tool's core capabilities.

4.1 Checking Device Drivers with Dxdiag

1. While on the *Win11-OS* system, click on the **Start** button and type **dxdiag** in the search box, then click on the **dxdiag** search result to launch the program.





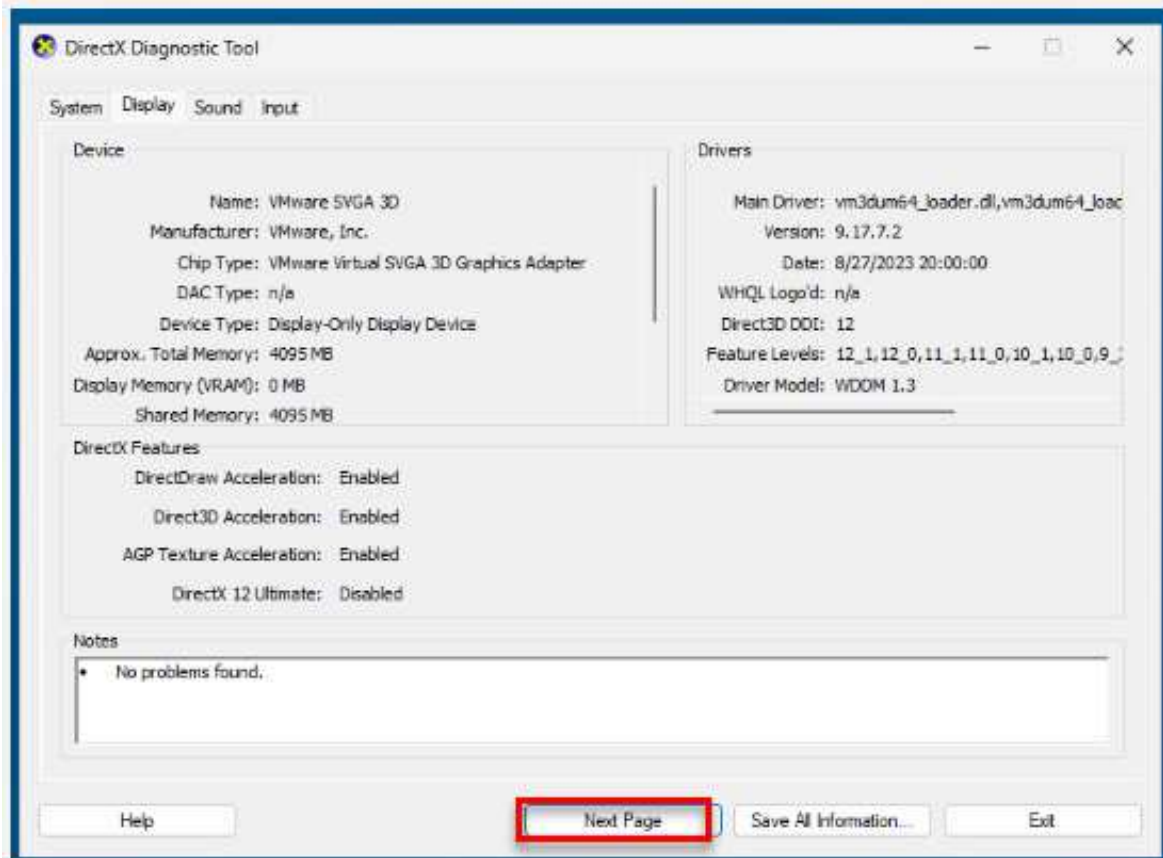
Alternatively, you can also access this utility by using the *Run* command window and typing the command **dxdiag.exe**.

2. Before the program opens, it will ask about checking to see if drivers are digitally signed. While this is usually a good practice, in this lab, you will click **No** to proceed.



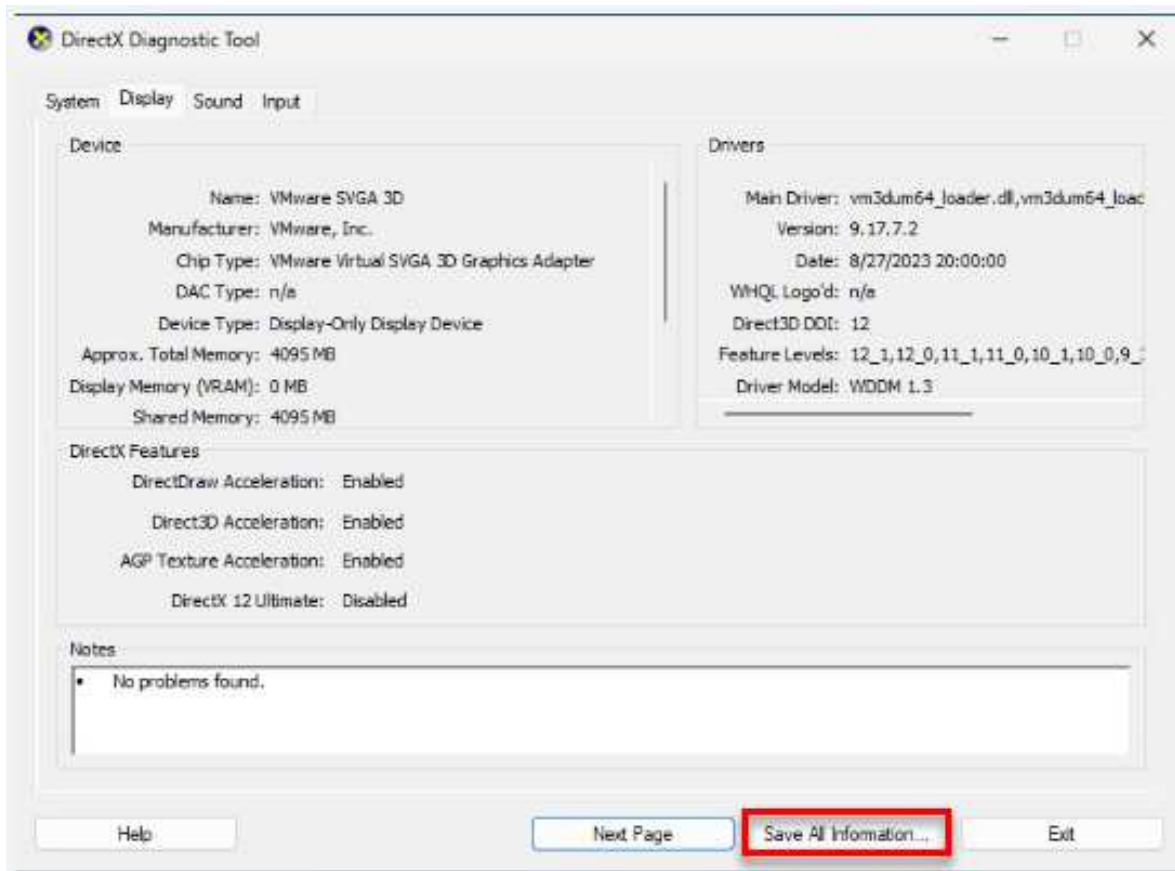
When the tool opens, you will be presented with four category tabs; *System*, *Display*, *Sound*, and *Input*. If there is more than one of any device in a category, each will be represented with its own tab.

- To become familiar with the information that can be found on each device, navigate through the tabs. Click the **Next Page** button to move through the tabs and review the types of information found on each. Notice on each page for every device there is a *Notes* pane at the bottom of the page. If problems exist, they are described here.

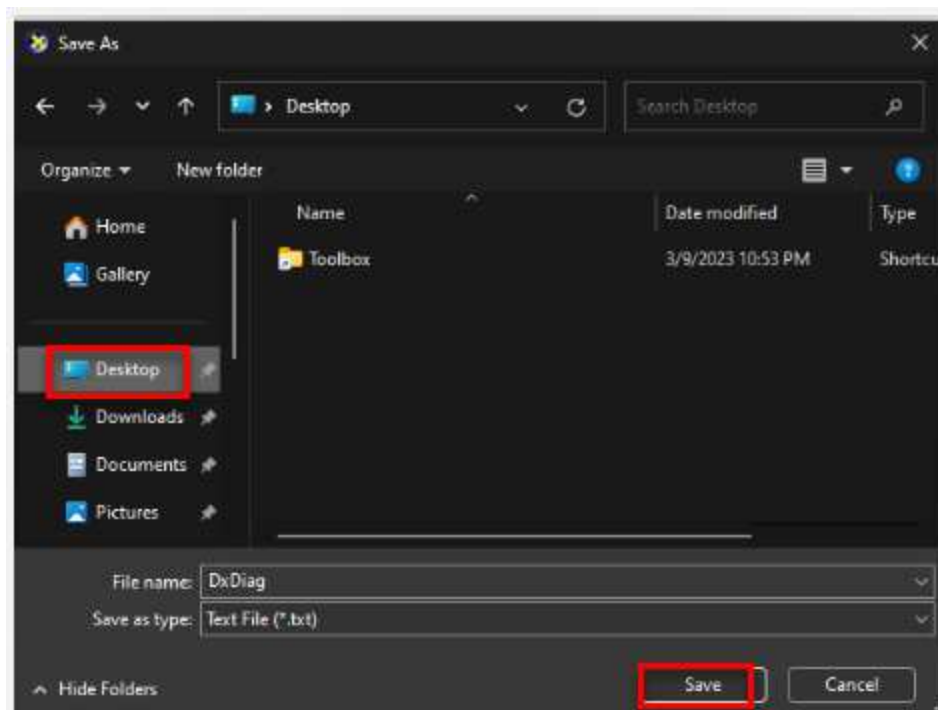


When you are troubleshooting system problems, there may be times when you work in collaboration with other technicians or vendors. Being able to share information is very important in order to work together productively. In the event that assistance is needed, using a diagnostic tool to collect system information and save settings can be very helpful.

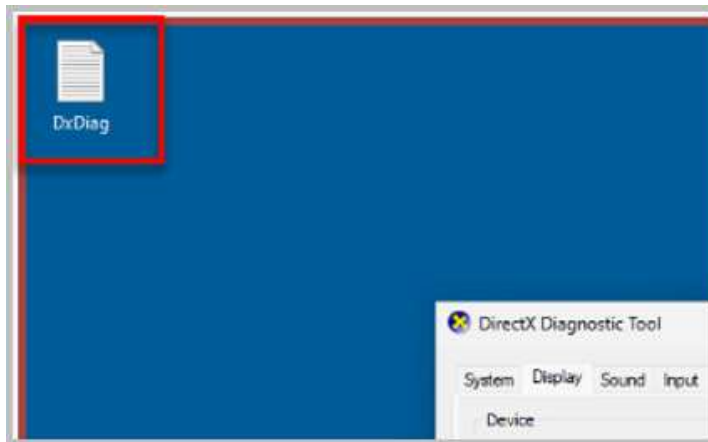
4. Click on the **Save All Information** button at the bottom of the page.



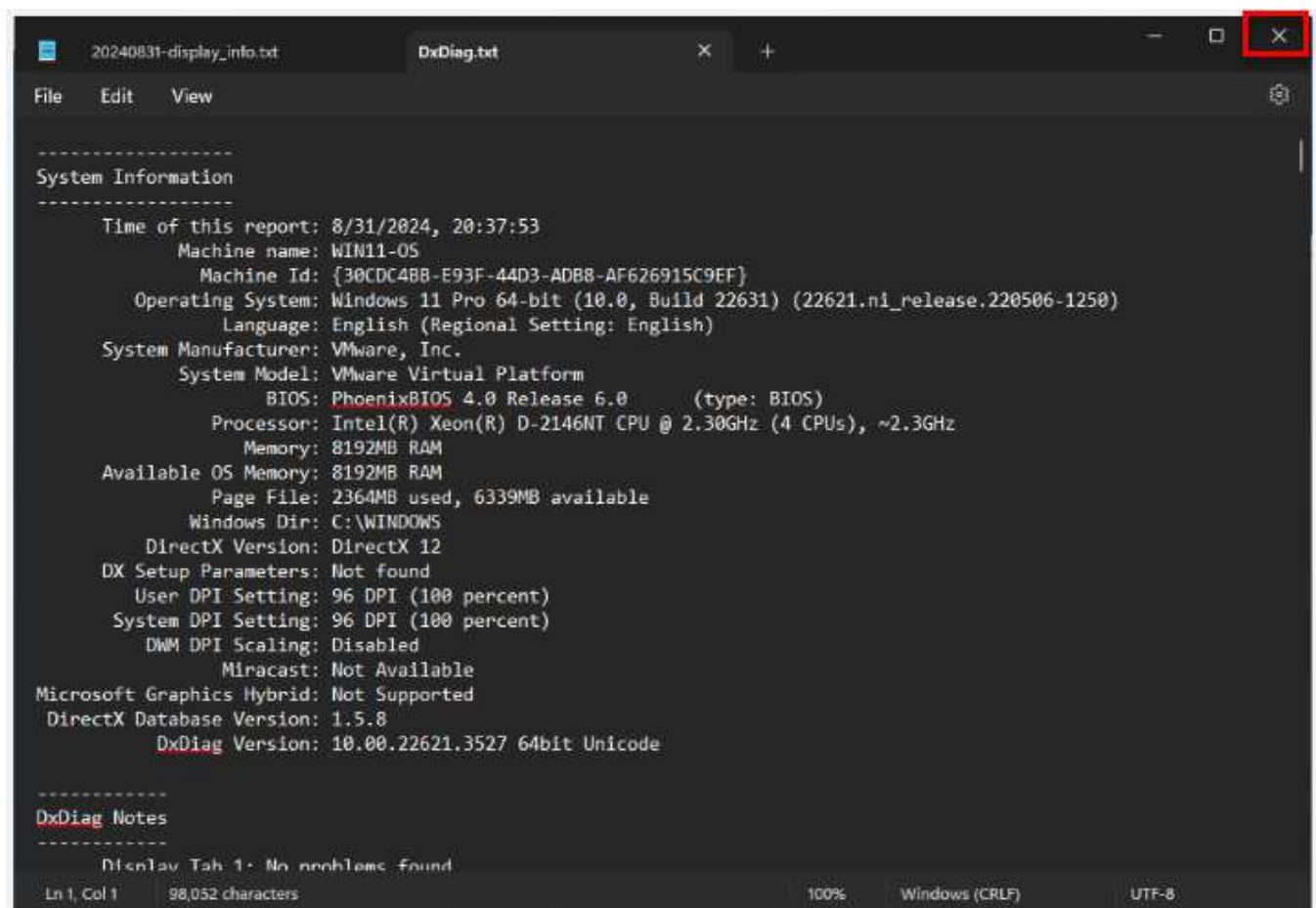
5. In the **Save As** window, click on **Desktop** in the left pane, and in the *File name* text field, leave the default filename of DxDiag. Click **Save**.



- Double-click the **DxDiag** file that is now on your *Desktop*. It contains all the information that was discovered when using the *DirectX Diagnostic Tool*



- Review the information and then close the *DxDiag - Notepad* window.

A screenshot of a Notepad window titled 'DxDiag.txt'. The window has a dark background and shows the output of the DirectX Diagnostic Tool. The text is as follows:

```
-----
System Information
-----
Time of this report: 8/31/2024, 20:37:53
Machine name: WIN11-05
Machine Id: {30CDC48B-E93F-44D3-ADB8-AF626915C9EF}
Operating System: Windows 11 Pro 64-bit (10.0, Build 22631) (22621.n1_release.220506-1250)
Language: English (Regional Setting: English)
System Manufacturer: VMware, Inc.
System Model: VMware Virtual Platform
BIOS: PhoenixBIOS 4.0 Release 6.0 (type: BIOS)
Processor: Intel(R) Xeon(R) D-2146NT CPU @ 2.30GHz (4 CPUs), ~2.3GHz
Memory: 8192MB RAM
Available OS Memory: 8192MB RAM
Page File: 2364MB used, 6339MB available
Windows Dir: C:\WINDOWS
DirectX Version: DirectX 12
DX Setup Parameters: Not found
User DPI Setting: 96 DPI (100 percent)
System DPI Setting: 96 DPI (100 percent)
DWM DPI Scaling: Disabled
Miracast: Not Available
Microsoft Graphics Hybrid: Not Supported
DirectX Database Version: 1.5.8
DxDiag Version: 10.00.22621.3527 64bit Unicode

-----
DxDiag Notes
-----
Display Tab 1: No problems found
```

The status bar at the bottom shows 'Ln 1, Col 1', '98,052 characters', '100%', 'Windows (CRLF)', and 'UTF-8'. The window's title bar includes a red box around the close button.

- Close any remaining windows open on the *Win11* machine.
- The lab is now complete; you may end the reservation.